

Exterior Winding-Zero Does Not Force Closedness for Differential Chains

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Source Problem

The source is Harrison Pugh, *Applications of Differential Chains to Complex Analysis and Dynamics*, arXiv:1012.5542 [1]. Remark 3.7.2 follows the generalized Cauchy residue theorem and proposes that the closedness assumption may be unnecessary:

ν_k . Thus, $\text{supp}(\nu) = \bigcup_k \text{supp}(\nu_k) = \text{supp}(\nu) \setminus (\bigcup_k \text{supp}(\nu_k))$. Therefore, J is holomorphic on a neighbourhood of $\text{supp} K = \sum_k \text{Ind}_J(a_k) D'_k$. By Theorem 3.5.1,

$$\oint_{\partial(K - \sum_k \text{Ind}_J(a_k) D'_k)} f(z) dz = 0.$$

Therefore,

$$\oint_J f(z) dz = \oint_{\partial K} f(z) dz = \oint_{\partial(\sum_k \text{Ind}_J(a_k) D'_k)} f(z) dz = \sum_k \text{Ind}_J(a_k) \oint_{B'_k} f(z) dz.$$

□

Remark 3.7.2. We conjecture that the requirement that J be closed in Theorem 3.7.1 is unnecessary. The requirement that $\text{Ind}_J(w) = 0$ for all $w \in U^c$ is very strong and should imply that $\partial J = 0$. Moreover, the requirement that K have finite mass could also probably be dropped. We only need it so that signed density

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The key sentence for this packet is the proposed implication that

$$\text{Ind}_J(w) = 0 \quad (w \in U^c) \quad \text{should imply} \quad \partial J = 0.$$

We give a compactly supported differential-chain counterexample to exactly that implication.

Result

Theorem 1. *There are a bounded open set $U \subset \mathbb{C}$ and a compactly supported differential 1-chain $J \in \hat{\mathcal{B}}_1^1(\mathbb{R}^2)$, with $\text{supp } J \subset U$, such that*

$$\text{Ind}_J(w) = 0 \quad \text{for every } w \in U^c,$$

but J is not closed:

$$\partial J \neq 0.$$

Consequently, the implication proposed in Remark 3.7.2 of the source paper is false.

Proof Intuition

The winding number only tests J against the Cauchy kernels $(z - w)^{-1} dz$, and for $w \notin U$ these are holomorphic near the support of the chain. A $\partial_{\bar{z}}$ -exact density is invisible to all such holomorphic tests by integration by parts. But the same chain can still have a real boundary, because closedness tests J against all exact real 1-forms, not only holomorphic differentials.

Proof

Proof. Let $U \subset \mathbb{C}$ be a bounded open disk. Choose a real-valued, nonconstant bump function $\phi \in C_c^\infty(U)$. Define the smooth compactly supported vector field

$$X = \frac{1}{2} \left(\frac{\partial \phi}{\partial x} \partial_x + \frac{\partial \phi}{\partial y} \partial_y \right).$$

Let J be the differential 1-chain represented by X against planar Lebesgue measure dA :

$$\int_J \omega = \int_U \iota_X \omega dA$$

for every test 1-form ω . This is the local compact-support version of the source's vector-field construction of chains $\xi_{X,\mu}$, which places J in $\hat{\mathcal{B}}_1^1$.

Write $z = x + iy$. Since $dz(X) = \frac{1}{2}(\phi_x + i\phi_y) = \partial_{\bar{z}}\phi$, for every $w \in U^c$ we have

$$\int_J \frac{dz}{z - w} = \int_U \frac{\partial_{\bar{z}}\phi(z)}{z - w} dA(z).$$

The function $z \mapsto (z - w)^{-1}$ is holomorphic on a neighborhood of $\text{supp } \phi$. Therefore

$$\frac{\partial_{\bar{z}}\phi(z)}{z - w} = \partial_{\bar{z}} \left(\frac{\phi(z)}{z - w} \right).$$

Because ϕ has compact support in U , the last function is compactly supported in the plane. Hence

$$\int_U \partial_{\bar{z}} \left(\frac{\phi(z)}{z - w} \right) dA(z) = 0,$$

by applying Stokes' theorem, or simply by integrating the compactly supported first partial derivatives over $\mathbb{R}^2$. It follows that

$$\text{Ind}_J(w) = \frac{1}{2\pi i} \int_J \frac{dz}{z - w} = 0 \quad (w \in U^c).$$

Now test the boundary of J against the smooth 0-form ϕ . Stokes' formula for differential chains gives

$$\int_{\partial J} \phi = \int_J d\phi = \int_U d\phi(X) dA = \frac{1}{2} \int_U |\nabla \phi|^2 dA.$$

Since ϕ is not constant and has compact support, this integral is strictly positive. Therefore $\partial J \neq 0$. This proves the claimed counterexample. $\square$

What This Does and Does Not Settle

The counterexample disproves the specific mechanism proposed in Remark 3.7.2: vanishing exterior winding number does not force a differential 1-chain to be closed. It does not, by itself, disprove the residue theorem after removing closedness. Also note that if the theorem’s current assumption $\partial K = J$ is retained, then closedness follows automatically from $\partial^2 K = 0$. Thus the remaining meaningful issue is not this implication but whether the formula can be repaired or extended under a different hypothesis, and whether the finite-mass requirement on K can be weakened.

Additional Diagnostic: Cauchy Formula Needs Closedness

The thesis also states a generalized Cauchy integral formula without an explicit closedness assumption on J , and the later paper *Generalizations of the Cauchy Integral Theorems* repeats the same formulation [2]. The proof invokes the generalized Cauchy integral theorem, which does require $\partial J = 0$. A classical segment already shows that some such assumption is necessary.

Indeed, let J be the oriented line segment from a to b , with $a \neq b$, and let $z \notin [a, b]$. Put $f(\zeta) = \zeta - z$. Then $f(z) = 0$, so the left side of the stated formula is 0. But

$$\frac{1}{2\pi i} \int_J \frac{f(\zeta)}{\zeta - z} d\zeta = \frac{1}{2\pi i} \int_J 1 d\zeta = \frac{b - a}{2\pi i} \neq 0.$$

This diagnostic is separate from Theorem 1; it is recorded here because it points to the same missing closedness issue.

Novelty and Verification Notes

The local run registry and cheap text indexes were searched for arXiv:1012.5542 and for the phrases “Ind.J(w)”, “partial J”, “closed”, and “differential chains”. The only local nearby result concerned a different Banach flat-chain target, arXiv:2508.08017. Web searches on 2026-07-18 for the exact Remark 3.7.2 wording and close variants found the source thesis and the later arXiv:1101.0385 paper, but no known counterexample to the implication handled here.

For human review, the key technical point is membership of the vector-field chain $J = \xi_{X,dA}$ in the intended local differential-chain class. The source develops such chains from vector fields and Radon measures; with smooth compact support this is the least singular case of that construction.

References

- [1] Harrison Pugh, *Applications of Differential Chains to Complex Analysis and Dynamics*, arXiv:1012.5542, 2010.
- [2] Jenny Harrison and Harrison Pugh, *Generalizations of the Cauchy Integral Theorems*, arXiv:1101.0385, 2011.